

Unit-3 - Ring

* Ring: An algebraic system $(R, +, \cdot)$ is called a ring if binary operations $+$ & $\cdot$ on R satisfies the following properties.

- ① $(R, +)$ is an abelian group
- ② $(R, \cdot)$ is a semi-group
- ③ Operation $(\cdot)$ is distributive over $+$.

In other words for any $a, b, c \in R$,

i) $a \cdot (b+c) = a \cdot b + a \cdot c$

ii) $(b+c) \cdot a = b \cdot a + c \cdot a$

Examples of ring:

- ① Set of all integers is a ring with unity for $+$ & $\cdot$.
- ② Set of all real No. of form $a + \sqrt{2}b$ is a ring for $+$ & $\cdot$.
- ③ Set of all matrices of form $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is a ring for $+$ & $\cdot$.
- ④ Set of complex numbers with usual addition & multiplication is a ring.

* Commutative Ring: A ring is said to be commutative ring, if $a \cdot b = b \cdot a, \forall a, b \in R$.

Examples of commutative Ring:

- ① $(\mathbb{Z}, +, \cdot)$ is a commutative ring with unity.
- ② $(2\mathbb{Z}, +, \cdot)$ is a commutative ring without unity where $2\mathbb{Z} =$ set of even integers.

③ Set of $n \times n$ matrix over real numbers w.r.t. usual matrix addition & multiplication is a non-commutative ring with unity.

④ Ring with unity: A ring $(R, +, \cdot)$ is said to be ring with unity if $\forall a \in R$ there exist $1 \in R$ such that $a \cdot 1 = 1 \cdot a = a$

* Properties of Ring: If $(R, +, \cdot)$ is a ring with identity 0 & unit element 1 then following are true for all $a, b, c \in R$

i) $a \cdot 0 = 0 \cdot a = 0$

ii) $a \cdot (-b) = (-a) \cdot b = -(a \cdot b)$

iii) $(-a) \cdot (-b) = a \cdot b$

iv) Unit element is unique.

* Sub-Ring: Let $(R, +, \cdot)$ be a ring. A non-empty sub-set S of R is said to be sub-ring of R if $(S, +, \cdot)$ is a ring.

Ex. — $(\mathbb{Z}, +, \cdot)$ is a subring of $(\mathbb{R}, +, \cdot)$

* Zero divisors: Let $(R, +, \cdot)$ is a commutative ring. And element $a \neq 0$ in R is said to be 0 divisor if there exist $b \neq 0$ in R , such that $a \cdot b = 0$.

* A ring $(R, +, \cdot)$ is said to be without zero divisor if $a \cdot b = 0 \Rightarrow a = 0$ OR $b = 0$, $\forall a, b \in R$.

* Integral Domain :

A commutative ring $(R, +, \cdot)$ with identity 1 without zero divisor is called an integral domain. If $a \cdot b = 0 \Rightarrow a = 0$ OR $b = 0, \forall a, b \in R$ then it is a ring without zero divisor.

Ex. - ① $(\mathbb{Z}, +, \cdot), (\mathbb{Z}, +, \cdot), (\mathbb{Q}, +, \cdot), (\mathbb{R}, +, \cdot)$ are integral domain.

② $(\mathbb{Z}_4, +, \cdot)$ is a ring with zero divisor so, it is not integral domain.

* Field : A commutative ring with unity in which every non-zero elements possesses their multiplicative inverse, is called as field.

* Note - A field is an integral domain.

Ex. - ① $(\mathbb{R}, +, \cdot), (\mathbb{C}, +, \cdot), (\mathbb{Q}, +, \cdot)$ are fields.

② $(\mathbb{Z}, +, \cdot)$ is integral domain but not field.

Q. Prove that $\mathbb{Z}_4$ is a ring under addⁿ & multiplication modulo 4. Addⁿ & multiplication table for $\mathbb{Z}_4$

→ Let $\mathbb{Z}_4 = \{0, 1, 2, 3\}$

+	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

x	0	1	2	3
0	0	0	0	0
1	0	1	2	3
2	0	2	0	2
3	0	3	2	1

① $(\mathbb{R}, +)$ is an abelian group,

2) Closure: $a * b \in \mathbb{Z}_4$ i.e. $\forall a + b \in \mathbb{Z}_4$

Ex. — $1 + 2 = 3 \in \mathbb{Z}_4$

$\therefore$ It is closure.

b) Associative: $a + (b + c) = (a + b) + c \quad \forall a, b, c \in \mathbb{Z}_4$

$$0 +_4 (1 +_4 2) = (0 +_4 1) +_4 2$$

$$3 = 3 \in \mathbb{Z}_4$$

$\therefore$ It is associative.

c) Identity:

$$a + e = e + a = a$$

$$0 + 0 = 0$$

$$1 + 0 = 1$$

$$2 + 0 = 2$$

$$3 + 0 = 3$$

$\therefore$ It is an identity.

$\therefore$ Identity element is 1.

d) Inverse element: $a + b = b + a = e$

from table:

$$(0)^{-1} = 0 \quad (3)^{-1} = 3$$

$$(1)^{-1} = 3$$

$$(2)^{-1} = 2$$

e) Commutative: $a + b = b + a, \quad \forall a, b \in \mathbb{Z}_4$

$$1 +_4 2 = 2 +_4 1$$

$$3 = 3$$

$\therefore$ It's commutative.

So, all cond's are satisfied.

$\therefore (\mathbb{R}, +)$ is an abelian group.

* Semi-group :

$(R, \cdot)$

a) Closed - $\forall a, b \in \mathbb{Z}_4$

$$1 \cdot 2 = 2 \in \mathbb{Z}_4$$

$\therefore$ It is closed.

b) Associative - $a \cdot (b \cdot c) = (a \cdot b) \cdot c$

$$\therefore 1 \cdot (2 \cdot 3) = (1 \cdot 2) \cdot 3$$

$$\therefore 2 = 2$$

$\therefore$ It is associative.

~~Id~~

$\therefore$ It is semi-group.

③ i) $a \cdot (b+c) = a \cdot b + a \cdot c$

$$\left. \begin{array}{l} \text{L.H.S.} = 1 \cdot (2+3) = 1 \cdot (1) = 1 \\ \text{R.H.S.} = 1 \cdot 2 + 1 \cdot 3 = 2+3 = 1 \end{array} \right\} \text{L.H.S.} = \text{R.H.S.}$$

ii) $(b+c) \cdot a = b \cdot a + c \cdot a$

$$\left. \begin{array}{l} \therefore \text{L.H.S.} = (2+3) \cdot 1 = 1 \\ \text{R.H.S.} = 2 \cdot 1 + 3 \cdot 1 \\ = 2+3 = 1 \end{array} \right\} \text{L.H.S.} = \text{R.H.S.}$$

$\therefore$ It is distributive.

$\therefore$ All properties are satisfied.

$\therefore (\mathbb{Z}_4, +, \cdot)$ is a ring.

Q. $R = \{0, 2, 4, 6, 8\}$ is a commutative ring under add^n & multiplication modulo 10.

Q. Let $M = \left\{ \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix} / a \in R \right\}$ & let $+$ & $\cdot$ denote matrix add^n & multiplication. Is $(M, +, \cdot)$ an integral domain?

→ For commutative ring,

$$A \times B = B \times A \quad \forall A, B \in M$$

$$\text{Let } A = \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix} \text{ & } B = \begin{bmatrix} b & 0 \\ 0 & b \end{bmatrix}$$

$$\text{Let } A \times B = B \times A$$

$$\begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix} \begin{bmatrix} b & 0 \\ 0 & b \end{bmatrix} = \begin{bmatrix} b & 0 \\ 0 & b \end{bmatrix} \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix}$$

$$\begin{bmatrix} ab & 0 \\ 0 & ab \end{bmatrix} = \begin{bmatrix} ba & 0 \\ 0 & ba \end{bmatrix}$$

$$\begin{bmatrix} ab & 0 \\ 0 & ab \end{bmatrix} = \begin{bmatrix} ab & 0 \\ 0 & ab \end{bmatrix} \quad \therefore ab = ba.$$

* For add^n :

$$A + B = B + A$$

$$\begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix} + \begin{bmatrix} b & 0 \\ 0 & b \end{bmatrix} = \begin{bmatrix} b & 0 \\ 0 & b \end{bmatrix} + \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix}$$

$$\begin{bmatrix} a+b & 0 \\ 0 & a+b \end{bmatrix} = \begin{bmatrix} b+a & 0 \\ 0 & b+a \end{bmatrix}$$

$$\begin{bmatrix} a+b & 0 \\ 0 & a+b \end{bmatrix} = \begin{bmatrix} a+b & 0 \\ 0 & a+b \end{bmatrix}$$

$(M, +, \times)$ is a commutative ring.

For unity:

$$A \times I = I \times A = A$$

$$\begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix}$$

i.e. $A \times I = A$

where $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \text{Identity}$

$(R, +, \times)$ is a commutative ring with identity.

Let $AB = 0$

$$\begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix} \times \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$\Rightarrow A = 0 \text{ OR } B = 0$

It is without zero divisor.

$(M, +, \times)$ is a commutative ring with identity ~~without~~ zero divisor.

$(M, +, \times)$ is an ~~int~~ integral domain.

Q. Give example of following.

- 1) A commutative ring with identity
- 2) A non-commutative ring with identity
- 3) Neither commutative nor as unique element.
- 4) Integral domain but not a field.

→ 1) Set of integers.

2) Set of all $n \times n$ matrices with all entries under $+$ & $\times$ of matrix.

3) Set of all $n \times n$ matrices with element of form $2k \text{ & } k \in \mathbb{Z}$ under $+$ & $\times$.

4) i) $S = \{a + b\sqrt{2} ; a, b \in \mathbb{Z}\}$ is integral domain but not a field.

ii) $(\mathbb{Z}, +, \times)$ - Integral domain but not field.

Q. Show that $S = \{a + b\sqrt{2} ; a, b \in \mathbb{Z}\}$ is integral domain but not field.

* Partial Ordered Set (Poset): A relation R on a set A is called Partial ordered relation if R is reflexive:
 $aRa \quad \forall a \in R$

2) Anti-symmetric - aRb, bRa then $a = b$

3) Transitive - If aRb, bRc then aRc .

A set 'A' together with partial order relation R is called Partial ordered set.

It is denoted by (A, R) OR $(A, \leq)$ where $\leq$ is a partial ordered relation.

Q. If S is any set & $P(S)$ is its power set (collection of all subsets of S) then relation $R \subseteq$ i.e. aRb iff $a \subseteq b$ is partial order relation on $P(S)$.

→ Let $a, b, c \in P(S)$

① Reflexive - If $aRa \forall a \in P(S)$
 $\therefore a \subseteq a$
 $\therefore R$ is reflexive.

② Anti-symmetric - If aRb, bRa then $a=b$
 $\therefore a \subseteq b, b \subseteq a$ then $a=b$
 $\therefore R$ is anti-symmetric.

③ Transitive - If aRb, bRc then aRc ,
 $\therefore a \subseteq b, b \subseteq c$ then $a \subseteq c$
 $\therefore R$ is transitive.

$\therefore$ All condⁿ are satisfied.

$\therefore R$ is partial ordered relation.

$\therefore (P(S), \subseteq)$ is a poset.

Q. If $\mathbb{Z}^+$ is set of positive integers then prove that $(\mathbb{Z}^+, \leq)$ is poset &

→

Let aRb iff $a \leq b$

① Reflexive - $aRa, a \leq a \forall a \in \mathbb{Z}^+$

$\therefore a \leq a$

$\therefore R$ is reflexive.

② Anti-symmetric - aRb, bRa then $a=b$
 $a \leq b, b \leq a$ then $a=b$
 $\therefore$ ~~It an~~ R is anti-symmetric

③ Transitive : aRb, bRc then aRc
 $a \leq b, b \leq c$ then $a \leq c$

$\therefore R$ is transitive.

$\therefore$ All condⁿs are satisfied

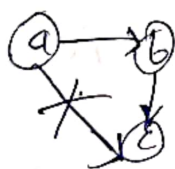
$\therefore R$ is partial ordered relation.

$(\mathbb{Z}^+, \leq)$ is a poset.

* Hasse Diagram:

Step - 1 - Delete loop at each vertex.

Step - 2 - Delete edges implied by transitivity.

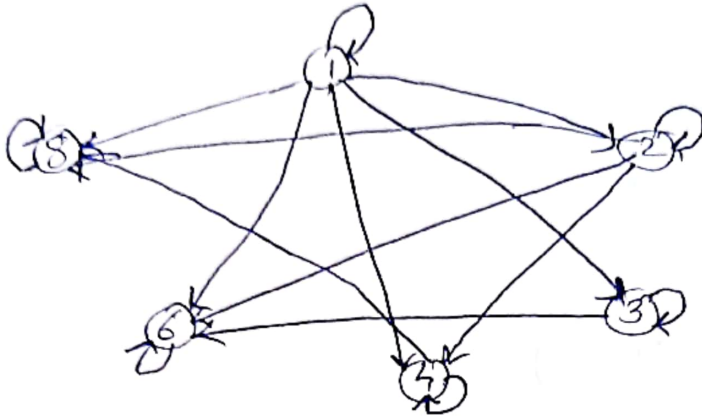


Step - 3 - Rearrange diagraph, if necessary such that all arrows goes upward direction.

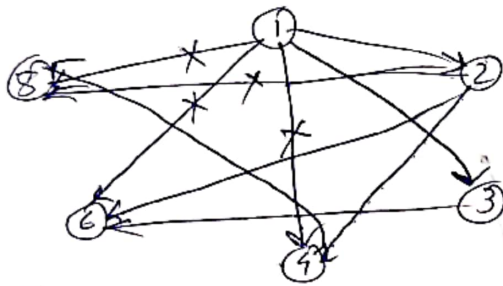
Step - 4 - Replace all arrows by edges & vertices by dot.

Q. Construct Hasse diagram for poset $(A, |)$ ($|$ - divisibility)
 i.e. aRb iff $a|b$
 where $A = \{1, 2, 3, 4, 6, 8\}$

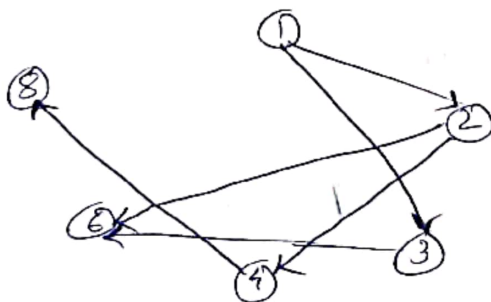
$$R = \{(1,1), (1,2), (1,3), (1,4), (1,6), (1,8), (2,2), (2,4), (2,6), (2,8), (3,3), (3,6), (4,4), (4,8), (6,6), (8,8)\}$$



Step - ① - Delete all loops -

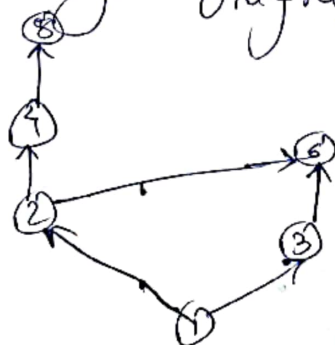


Step - ② - Delete transitive edges - $(1,2), (2,8) \rightarrow (1,8)$

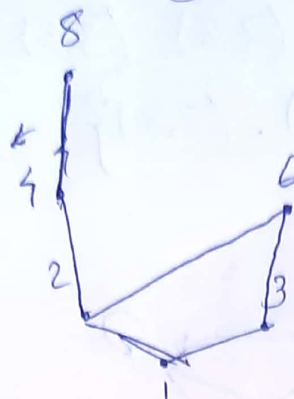


$$\begin{aligned} &(1,2), (2,6) \rightarrow (1,6) \\ &(1,2), (2,4) \rightarrow (1,4) \\ &(2,4), (2,8) \rightarrow (2,8) \end{aligned}$$

Step - ③ - Rearrange diagram in upward direction



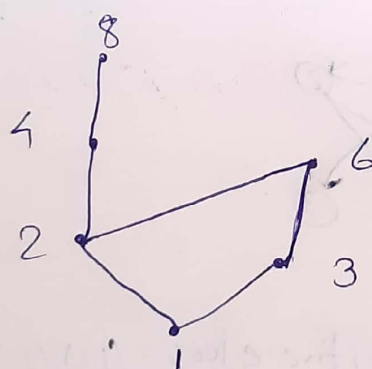
Step-1 - Replace arrows by edges & vertex by dot.



OR

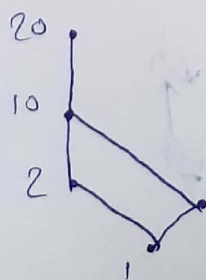
$$A = \{1, 2, 3, 4, 6, 8\}$$

Hasse diagram -

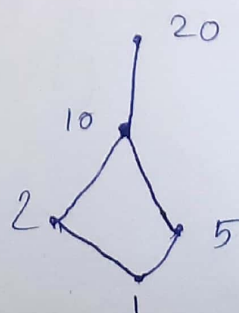


Q. Construct Hasse diagram of set $\{1, 2, 5, 10, 20\}$ under partial ordered relation divisibility i.e. $aRb \iff a|b$

$$R = \{(1, 1), (1, 2), (1, 5), (1, 10), (1, 20), (2, 2), (2, 10), (2, 20), (5, 5), (5, 10), (5, 20), (10, 10), (10, 20), (20, 20)\}$$

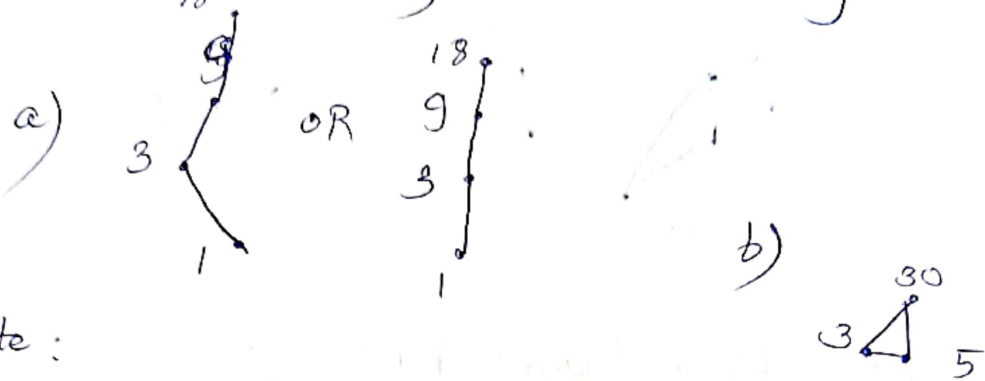


OR



Q. Draw Hasse diagram of following set under partial ordered relation divides.

a) $\{1, 3, 9, 18\}$ b) $\{3, 5, 30\}$



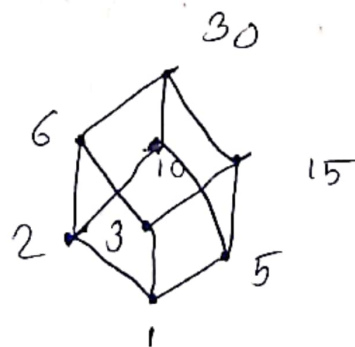
Note:

D_n = Set of positive divisors of n .

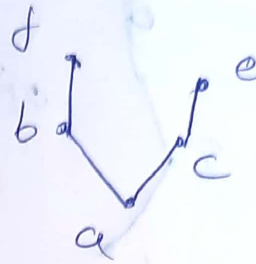
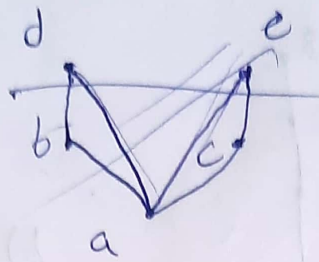
for D_n relation is always divisibility.
i.e. aRb iff $a|b$.

Q. Draw Hasse diagram of D_{30} .

→ $D_{30} = \{1, 2, 3, 5, 6, 10, 15, 30\}$
Hasse diagram.



Q. Let $A = \{a, b, c, d, e\}$ $R = \{(a, a), (b, b), (c, c), (d, d), (e, e), (a, b), (a, c), (a, d), (a, e), (b, d), (c, e)\}$
Draw Hasse diagram of (A, R)

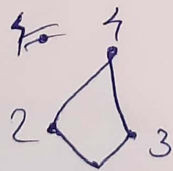


Q. Construct Hasse diagram of following relⁿ.

a) $A = \{1, 2, 3, 4\}$

$$R = \{(1,1), (1,2), (2,2), (2,4), (1,3), (3,3), (3,4), (1,4), (4,4)\}$$

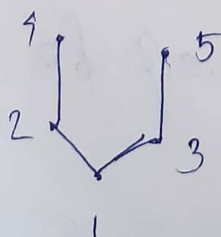
→



b) $A = \{1, 2, 3, 4, 5\}$

$$R = \{(1,1), (1,2), (1,3), (1,4), (1,5), (2,2), (2,4), (3,3), (3,5), (4,4), (5,5)\}$$

→



↑ Points to remember in Hasse Diagram:

1. It never has horizontal line.
2. Relation is always in upward direction.

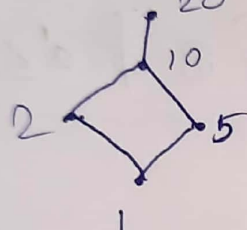
① Comparable element: If A is a given set & R is partial order relation on A , then elements a, b of A are said to be comparable if aRb OR bRa .

It means if aRb OR bRa , a & b aren't comparable.

$$\text{If } aRb \Rightarrow a \leq b$$

$$\& \text{ if } bRa \Rightarrow b \leq a$$

Ex. -



In this diagram 2 & 10 are comparable element also 1 & 10 , 1 & 20 , 2 & 10 , 5 & 20 are comparable but 2 & 5 aren't comparable.

② Maximal Elements: An element $a \in A$ is called maximal element of A if there is no element $c \in A$ such that $a \leq c$.

③ Minimal Element: An element $a \in A$ is called minimal element of A if there is no element $c \in A$ such that $c \leq a$.

④ Greatest Element: An element $a \in A$ is called greatest element of A if $x \leq a$, $\forall x \in A$.

⑤ Least Element: An element $a \in A$ is called least element of A if $a \leq x$, $\forall x \in A$.

⑥ Unit element: The greatest element of a poset if it exists, is denoted by I . & is called unit element.

⑦ Zero Element: The least element of poset, if it exists, is denoted by 0 (zero) ϕ is called zero element.

⑧ Upperbound: Consider a poset A ϕ a subset B of A . An element $a \in A$ is called an upperbound of B if $b \leq a, \forall b \in B$.

⑨ Lowerbound: An element $a \in A$ is called Lowerbound of B if $a \leq b, \forall b \in B$.

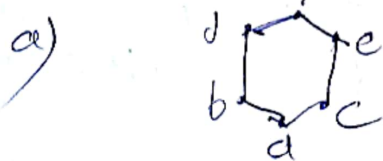
⑩ Greatest Lower Bound (GLB): An element $a \in A$ is called GLB of B if a is lower bound of B ϕ $a' \leq a$ whenever a' is lowerbound of B .

⑪ Least Upper Bound: An element $a \in A$ is called LUB of (LUB) B if a is upper bound of B ϕ $a \leq a'$ whenever a' is upperbound of B .

† Imp. Note:

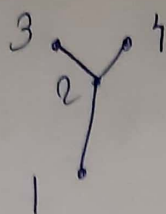
Maximal element ϕ minimal element may be more than one but greatest element ϕ least element are always unique if they exist.

Ex. — Determine least element, greatest element if it exists then find maximal element, minimal element of following poset.



least element = a
greatest element = f
maximal element = b, c
minimal element = a

b)



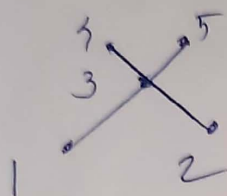
least element = 1

greatest element = ~~3, 4~~ doesn't define

maximal element = 3, 4

minimal element = 1

c)



least = No

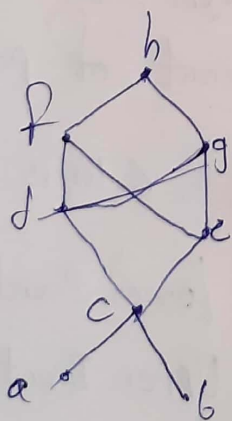
Max = 4, 5

greatest = No

Minimal = 1, 2

Q. Consider poset $A = \{a, b, c, d, e, f, g, h\}$ whose Hasse diagram is given below. Find all upper bounds, lower bounds, GLB, LUB. If it exists of following subset of A !

① $B_1 = \{a, b\}$ ② $B_2 = \{d, e\}$ ③ $B_3 = \{c, d, e\}$ ④ $B_4 = \{f, g, h\}$



→

① $B_1 = \{a, b\}$

Upper bounds = c, d, e, f, g, h

Lower bounds = doesn't exist

LUB

=

c

GLB

=

doesn't exist

$$\textcircled{2} B_2 = \{d, e\}$$

Upper = f, g, h

Lower = a, b, c

LUB = doesn't exist

GLB = c

$$\textcircled{3} B_3 = \{f, g, d\} \{c, d, e\}$$

Upper B. = f, g, h

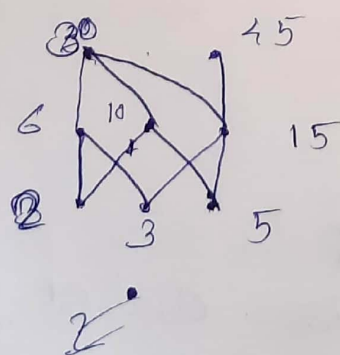
Lower B. = a, b, c

LUB = doesn't exist

GLB = c

Q. Find GLB, LUB, Upper bound of 2 & 5. Lower bound of 30 & 45. Also find maximal element, minimal element, greatest element & least element of poset (A, R) .

Here aRb iff $a|b$, $A = \{2, 3, 5, 6, 10, 15, 30, 45\}$



Lower Bound = $3, 5, 15$

Upper Bound = $10, 30$

LUB = 10

GLB = 15

Maximal elements = 30, 45

Minimal elements = 2, 3, 5

Greatest element = doesn't exist

Least element = doesn't exist

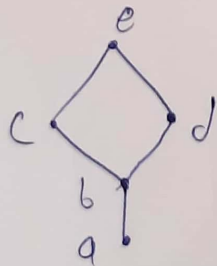
† Join($\vee$): Let $(A, \leq)$ be a poset & $a, b \in A$ then LUB of a & b is denoted by $a \vee b$, it is called join of a & b i.e. $a \vee b = \text{lub}(a, b)$.

† Meet($\wedge$): Let $(A, \leq)$ be a poset & $a, b \in A$ then GLB of a & b is denoted by $a \wedge b$. It is called meet of a & b .

† Lattice: Let $(L, \leq)$ be a poset, let a, b be two elements $\in L$ then L is said to be lattice if every pair of (a, b) of L has LUB & GLB.

† Note: ① If $a R a$, $a \vee a = a$ & $a \wedge a = a$
 ② If $a R b$, $a \vee b = b$ & $a \wedge b = a$

Q. Determine whether following poset is lattice or not?



→ Table
 $\text{Join}(\vee)$

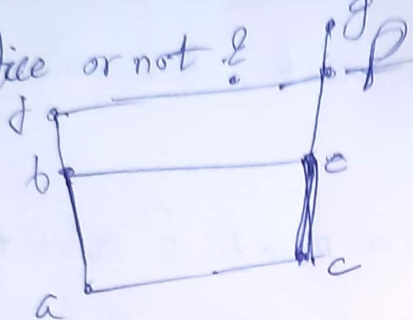
$\vee$	a	b	c	d	e
a	a	a	a	a	a
b	b	b	c	d	e
c	c	c	c	e	e
d	d	d	e	d	e
e	e	e	e	e	e

Meet Table($\wedge$)

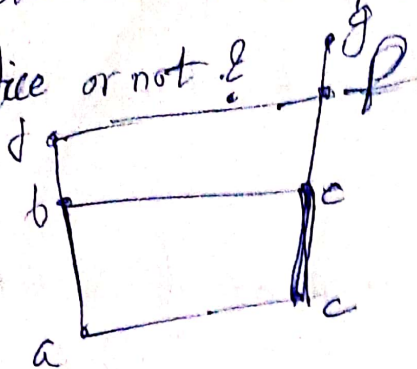
$\wedge$	a	b	c	d	e
a	a	a	a	a	a
b	a	b	b	b	b
c	a	b	c	c	c
d	a	b	c	d	d
e	a	b	c	d	e

Q. Determine whether poset with following Hasse diagram is

lattice or not?

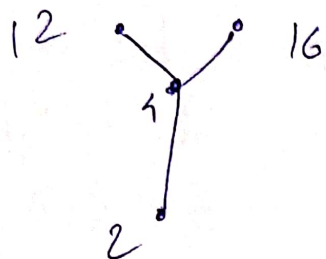


Q. Determine whether poset with following Hasse diagram is lattice or not?



→ It is a lattice

Q. Determine whether following poset is lattice?

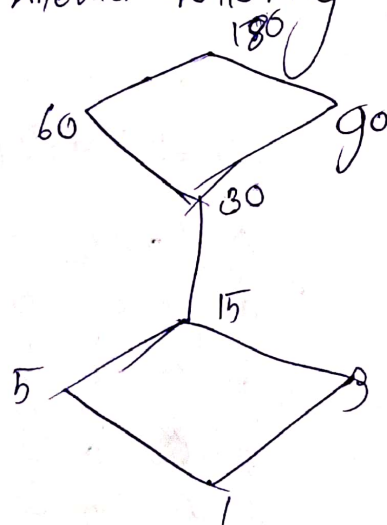


$\vee$	2	4	12	16
2	2	4	12	16
4	4	4	12	16
12	12	12	12	-
16	16	16	-	16

$12 \vee 16$ doesn't exist. So, it's not a lattice.

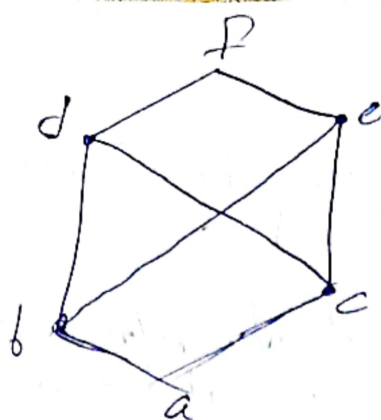
Q. Check whether following poset is lattice or not?

i)



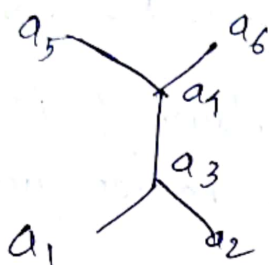
→ It is a lattice because each pair of element has LUB & GLB

(ii)



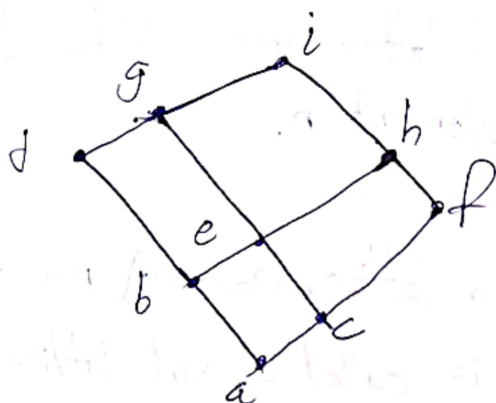
→ It isn't lattice because $b \vee c$ doesn't exist.

(iii)



→ It isn't lattice because $a_5 \vee a_6$ doesn't exist.

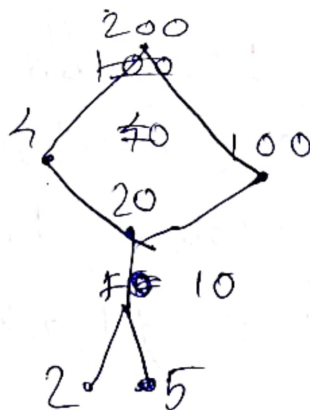
(iv)



→ It is a lattice, each pair of element has LUB & GLB.

Q. Is poset $A = \{2, 5, 10, 20, 40, 100, 200\}$ a lattice under relation divisible?

→ It's not lattice because $2 \wedge 5$ doesn't exist.



Q. Let $A = \{1, 2, 3, 4, 6, 9, 12\}$. Let a relation R on a set A is, $R = \{(a, b) / a \text{ divide } b, \forall a, b \in A\}$.

Give least of R prove that it is partial order relation. Draw Hasse diagram of set. Prove or disprove it is lattice?

Types of Lattices:

1) Bounded Lattice: A lattice L is said to be bounded if it has greatest element (1) & least element (0).

Ex. - 1) Lattice $\mathbb{Z}^+$ under partial order relation of divisibility is not bounded lattice since it has a least element the integer 1 but no greatest element.

2) Lattice $\mathbb{Z}$ under partial order relation $\leq$ is not bounded lattice because it has neither greatest element nor least element.

3) Lattice $\mathbb{D}_n$ is bounded lattice since it has least element 1 & greatest element n .

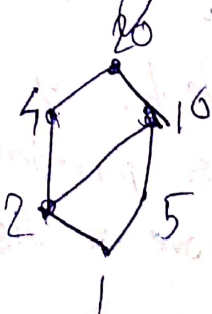
2) Sub-Lattice: Let $(L, \leq)$ be a lattice. A non-empty subset S of L is called a sub-lattice if for each $a \in S$ & $b \in S$ then $a \vee b \in S$ & $a \wedge b \in S$.

Ex - 1) Let $L = \mathbb{D}_{20}$ & $S = \{1, 4, 5, 20\}$. Check whether S is sublattice of $\mathbb{D}_{20}$ or not?

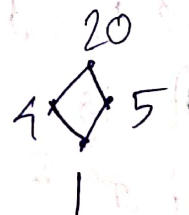
→ Let $L = \mathbb{D}_{20} = \{1, 2, 4, 5, 10, 20\}$
 $S = \{1, 4, 5, 20\}$

Hasse Diagram:

$\mathbb{D}_{20} =$



$S =$



V	1	4	5	20
1	1	4	5	20
4	1	4	20	20
5	5	20	5	20
20	20	20	20	20

V	1	4	5	20
1	1	1	1	1
4	1	4	1	4
5	1	1	5	5
20	1	4	5	20

$\therefore a \wedge b \nexists a \vee b$ exist, $\forall a, b \in S$
 $\therefore S$ is ~~not~~ sublattice of D_{20} .

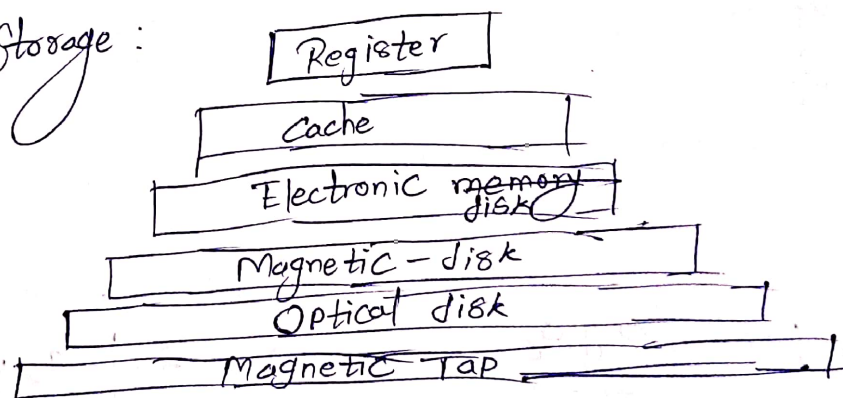
Unit $\leftarrow$ Pipelining

* I/O devices:

I/P - 1) keyboard 2) Mouse 3) Scanner

O/P - 1) Monitor 2) Printer 3) Speaker

* Mass Storage:

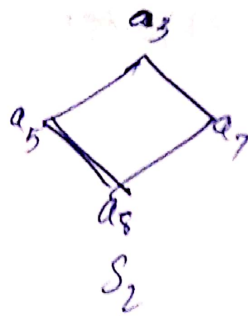
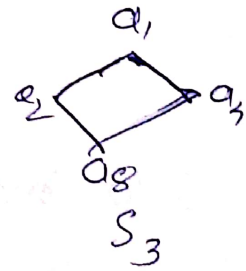
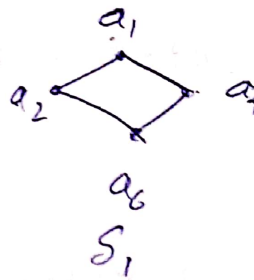
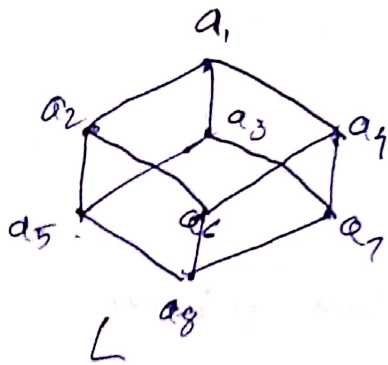


- i) Mass-storage devices is any storage device that make it possible to store large amount of data across computer, servers & IT environment.
- ii) These are portable storage medium that provides storage interface that can be internal & external to computer system.

$$L = \{a_1, a_2, \dots, a_8\} \quad \phi S_1 = \{a_1, a_2, a_4, a_6\}$$

$$S_2 = \{a_3, a_5, a_7, a_8\} \quad \phi S_3 = \{a_1, a_2, a_4, a_8\}$$

As shown in fig. examine whether S_1, S_2 & S_3 are sublattices



for S_1

v	a_1	a_2	a_4	a_6
a_1	a_1	a_1	a_1	a_1
a_2	a_1	a_2	a_4	a_2
a_4	a_1	a_4	a_4	a_4
a_6	a_1	a_2	a_4	a_6

1	a_1	a_2	a_4	a_6
a_1	a_1	a_2	a_4	a_6
a_2	a_2	a_2	a_6	a_6
a_4	a_4	a_6	a_4	a_6
a_6	a_6	a_6	a_2	a_6

$\therefore$ for $a \vee b$ & $a \wedge b \in S_1 \quad \forall a, b \in S_1$
 $\therefore S_1$ is sub-lattice of L .

for S_3 :

$\vee$	a_1	a_2	a_3	a_8
a_1	a_1	a_1	a_1	a_1
a_2	a_1	a_2	a_1	a_2
a_3	a_1	a_1	a_3	a_1
a_8	a_1	a_2	a_3	a_8

$\wedge$	a_1	a_2	a_3	a_8
a_1	a_1	a_2	a_3	a_8
a_2	a_2	a_2	a_6	a_8
a_3	a_1	a_6	a_3	a_8
a_8	a_8	a_8	a_8	a_8

$$\therefore a_2 \wedge a_3 = a_6 \notin S_3$$

$\therefore S_3$ isn't sub-lattice of L .

*Complemented lattice:

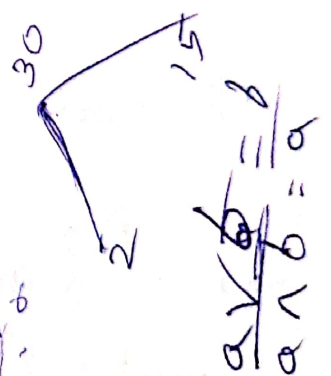
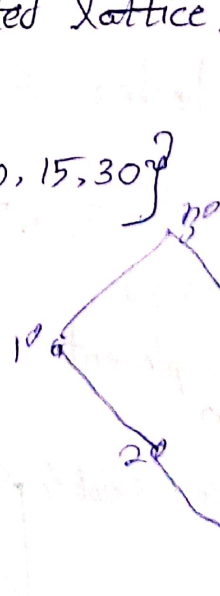
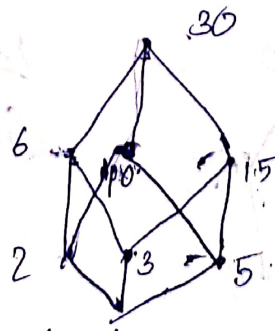
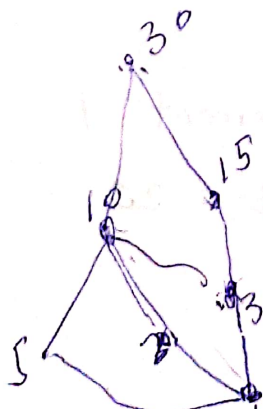
Let L be a bounded lattice. And element $\bar{a} \in L$ is said to be complement of a if $a \wedge \bar{a} = 0$ & $a \vee \bar{a} = 1$ (least element) (greatest element).

1 (greatest element).

A lattice L is said to be complemented lattice if there exist complement of every element of L .

Ex. — Check whether is complemented lattice.

$$\text{Let } D_{30} = \{1, 2, 3, 5, 6, 10, 15, 30\}$$



least element = 1
greatest element = 30

$$a \wedge \bar{a} = 1 \quad \& \quad a \vee \bar{a} = 30$$

element	Complements
1	30
2	15
3	10
5	6
6	5
10	3
15	2
30	1

$$a \wedge \bar{a} = 1 \quad \& \quad a \vee \bar{a} = 30$$

$$1 \wedge 30 = 1 \quad \& \quad 1 \vee 30 = 30$$

$$2 \wedge 15 = 1 \quad \& \quad 2 \vee 15 = 30$$

$\therefore \bar{a}$ exist, $\forall a \in D_{30}$

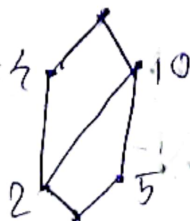
$\therefore D_{30}$ is complemented lattice.

Q. Check whether D_{20} is

$$D_{20} = \{1, 2, 4, 5, 10, 20\}$$

$$a \wedge \bar{a} = 1 \quad \& \quad a \vee \bar{a} = 20$$

Element	Complements
1	20
2	doesn't exist
4	5
5	4
10	doesn't exist
20	1



Least element = 1

Greatest = 20

$\&$ 10 complements don't exist.

$\therefore D_{20}$ isn't complemented lattice.

Q. Check following lattice is complemented or not.



→ least = a
greatest = e

$$a \wedge \bar{a} = a \quad \& \quad a \vee \bar{a} = e$$

Elements Complements

a	e
b	c
c	b, d
d	c
e	a

$\therefore \bar{a}$ exist $\forall a \in L$

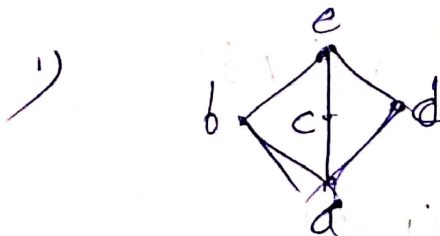
$\therefore$ It is complemented lattice.

④ Distributive Lattice: A lattice L is called distributive if for any element a, b, c in L . The following distributive properties are satisfied.

(i) $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$

(ii) $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$

Ex. - Examine following lattice is distributive.



$$\longrightarrow i) a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$$

$$a \wedge e = a \vee a$$

$$a = a$$

$$\therefore a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$$

$$ii) a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$$

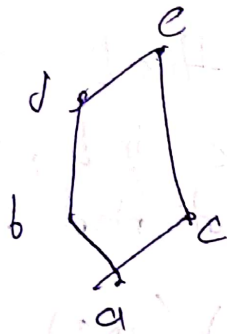
$$a \vee a = b \wedge c$$

$$a = a$$

$$\therefore a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$$

both condⁿs are satisfied. $\therefore$ It is distributive lattice.

2)



$$i) b \wedge (c \vee d) = (b \wedge c) \vee (b \wedge d)$$

$$b \wedge e = a \vee b$$

$$\therefore \overset{b}{b} \wedge (c \vee d) = (b \wedge c) \vee (b \wedge d)$$

$$ii) b \vee (c \wedge d) = (b \vee c) \wedge (b \vee d)$$

$$b \vee a = e \wedge d$$

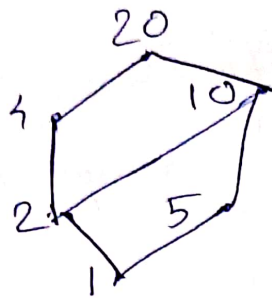
$$b \neq d$$

$$\therefore b \vee (c \wedge d) \neq (b \vee c) \wedge (b \vee d)$$

$\therefore$ It isn't distributive lattice.

Q. solve for D_{20} .

$$\rightarrow A = \{1, 2, 4, 5, 10, 20\}$$



$$\therefore i) 1 \wedge (2 \vee 5) = (1 \wedge 2) \vee (1 \wedge 5)$$

$$1 \wedge 10 = 1 \vee 1$$

$$1 = 1$$

$$ii) 1 \vee (2 \wedge 5) = (1 \vee 2) \wedge (1 \vee 5)$$

$$1 \vee 1 = 2 \wedge 5$$

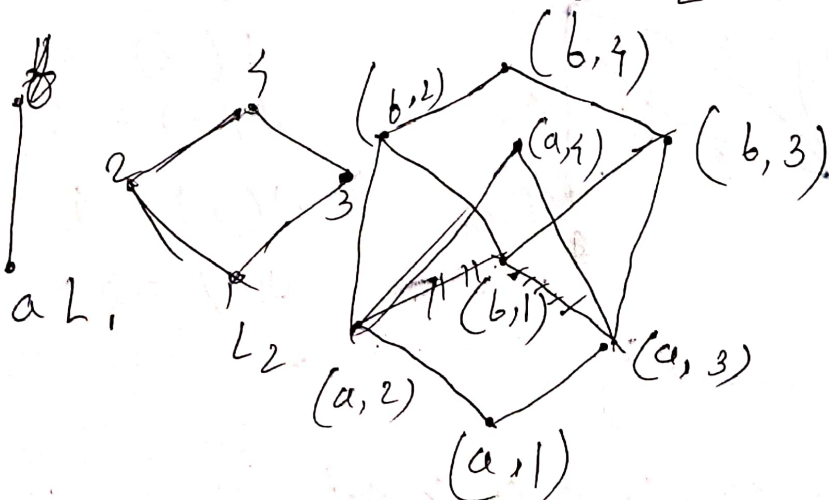
$$1 = 1$$

$\therefore$ It's distributive lattice.

Q. Check

* Direct Product of Lattice: If $(L_1, \leq)$ & $(L_2, \leq)$ are lattices then $L, \leq$ is a lattice where $L = L_1 \times L_2$ & partial ordered relation $\leq$ of L is product partial ordered.

Ex. - $L_1 \& L_2$, $L = L_1 \times L_2$





Unit-4 Graph Theory

† Graph: A graph is an ordered pair (V, E) of two sets $V \neq E$ satisfy following condⁿ:

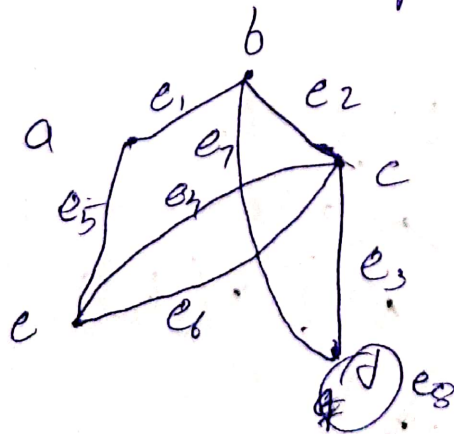
1) V is finite & non-empty.

2) Each $e \in E$ corresponds to a up unique unordered pair $v_1, v_2 \in V$

† Degree of Vertex: It is a number which indicates No. of edges incident to that vertex.

† Parallel Edges: If two edges have ~~set~~ same starting vertex & same any vertex then they are called parallel edges or multiple of edges.

† Loop: If an edge starts & ends in same vertex then it is called loop.



$$d(a) = 2$$

$$d(b) = 3$$

$$d(c) = 4$$

$$d(d) = 4$$

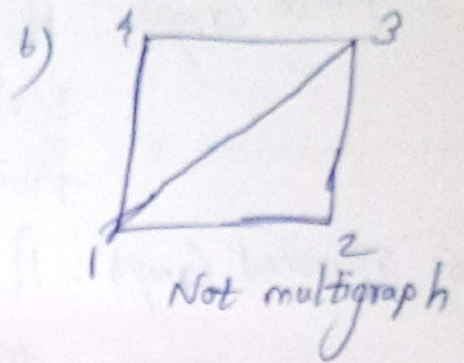
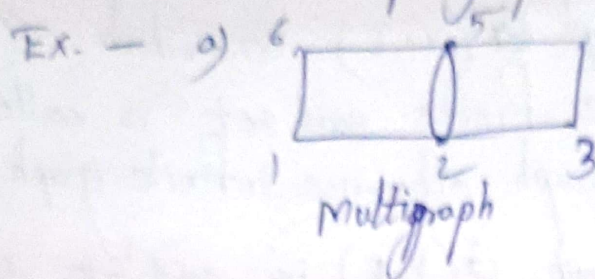
$$d(e) = 3$$

$$V = \{a, b, c, d, e\}$$

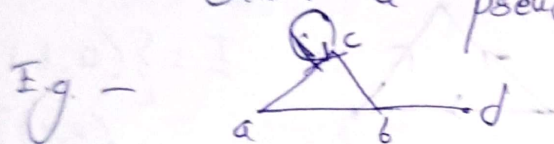
$$E = \{e_1, e_2, e_3, e_4, e_5, e_6, e_7\}$$

Types of Graphs:

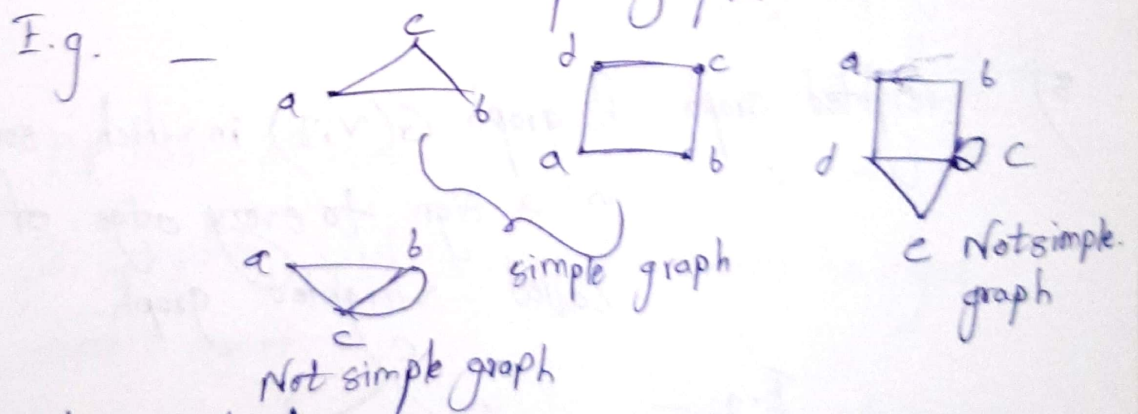
1) Multigraph: A graph in which a pair of vertices is joined by two or more edges is called multigraph or multiple graph.



2) Pseudograph: A graph having loops but no multiple edges is called a pseudograph.



3) Simple Graph: A graph without loop & multiple edges is called a simple graph.



Note:

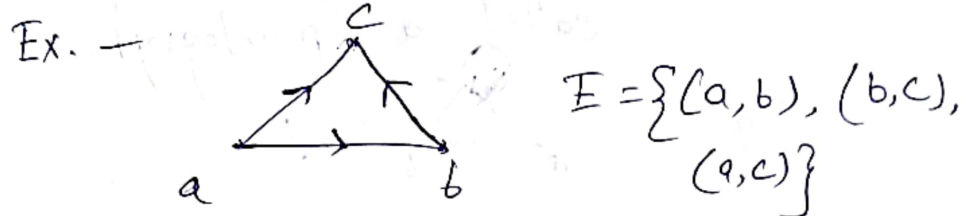
A simple graph having n vertices has maximum $\frac{n(n-1)}{2}$ no. of edges.

4) Null Graph: A graph $G(V, E)$ is said to be Null graph if E is an empty set. Null graph on n vertices is denoted by N_n .

Eg. - N_1 N_2 N_3 , N_n

5) Finite Graph: A graph $G(V, E)$ in which $V(x) \neq \emptyset$ & $E(x)$ are finite said set is called a finite graph otherwise infinite graph.

6) Directed Graph: A graph $G(V, E)$ is said to be directed graph if elements of E are an ordered pair of vertices.

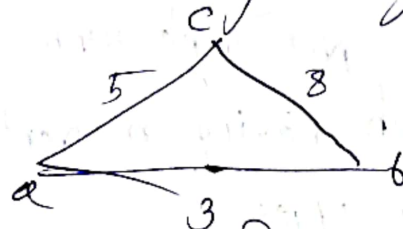


7) Non-directed:

A graph which is not directed is called non-directed graph.

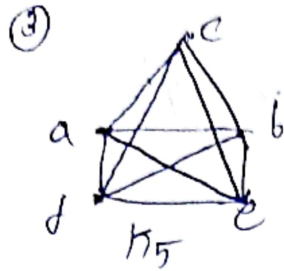
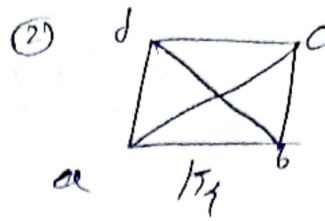
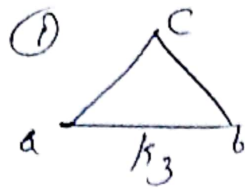
8) Weighted graph: A graph $G(V, E)$ in which some weight is a sign to every edge of G , is called weighted graph.

E.g. -



9) Complete graph: A simple graph of n vertices in which degree of each vertex is $(n-1)$ is called complete graph & is denoted by K_n .

E.g. -



Properties: 1) Every pair of vertices is adjacent in K_n .

2) In K_n each vertex has same degree as $(n-1)$.

3) Total No. of edges in K_n is $\frac{n(n-1)}{2}$.

E.g. - Can a complete graph with 8 vertices have 40 edges?

→ Complete with 8 vertices, No. of edges = $\frac{n(n-1)}{2}$

$$= \frac{8(8-1)}{2}$$

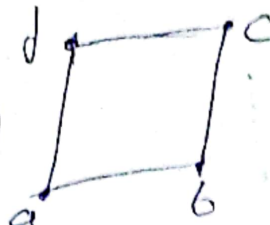
$$= 28$$

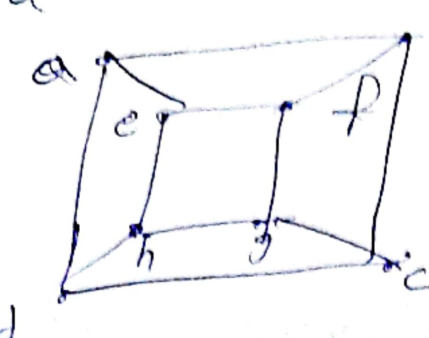
K_8 has 28 edges.

So, complete with 8 vertices having 40 edges doesn't exist.

* Regular graph: i) A graph in which every vertex has same degree is called a regular graph.

ii) If common degree of vertex is r , then it is called r regular graph. Therefore K_n is $(n-1)$ regular graph.

Ex. - 1)  2-regular graph.

2)  3-regular graph.

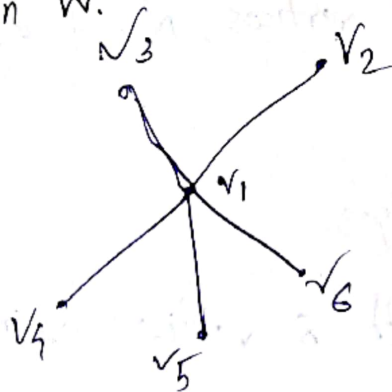
! Every complete graph is regular graph but converse is not true.

! Bipartite graph: A graph $G(V, E)$ is called bipartite

if: 1) V can be expressed as union of two disjoint set U & W . ($U \cup W = V$ & $U \cap W = \emptyset$)

2) Every edge in E has one vertex in U & other in W .

E.g. - 1)

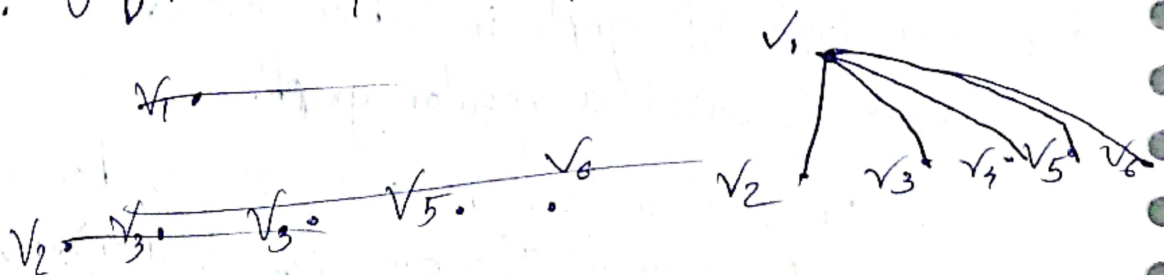


Let, $V = \{v_1, v_2, v_3, v_4, v_5, v_6\}$

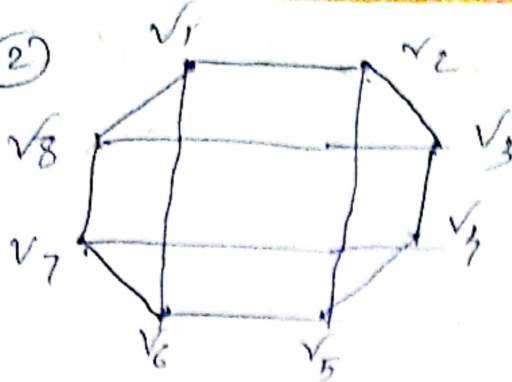
Let, $U = \{v_1\}$

$W = \{v_2, v_3, v_4, v_5, v_6\}$

$\therefore U \cup W = V$ & $U \cap W = \emptyset$



(2)

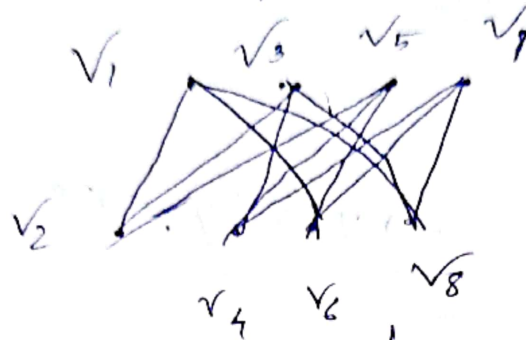


$$V = \{v_1, v_2, v_3, v_4, v_5, v_6, v_7, v_8\}$$

$$\text{Let } U = \{v_1, v_3, v_5, v_7\}$$

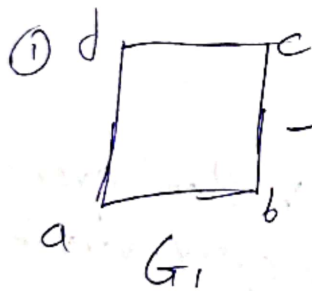
$$W = \{v_2, v_4, v_6, v_8\}$$

$$U \cup W = V \text{ \& } U \cap W = \emptyset$$



$\therefore G$ is bipartite graph.

Q. Determine which of following graphs are bipartite.



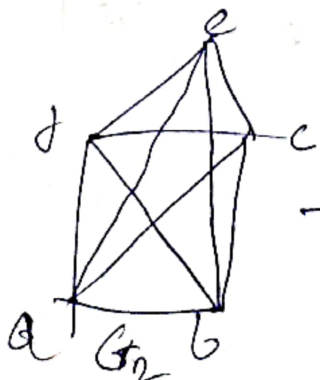
$$V = \{a, b, c, d\}$$

$$U = \{a, c\}, W = \{b, d\}$$

$$U \cup W = V \text{ \& } U \cap W = \emptyset$$



Q.



$$V = \{a, b, c, d, e\}$$

$$U = \{a, c, e\}, W = \{b, d\}$$

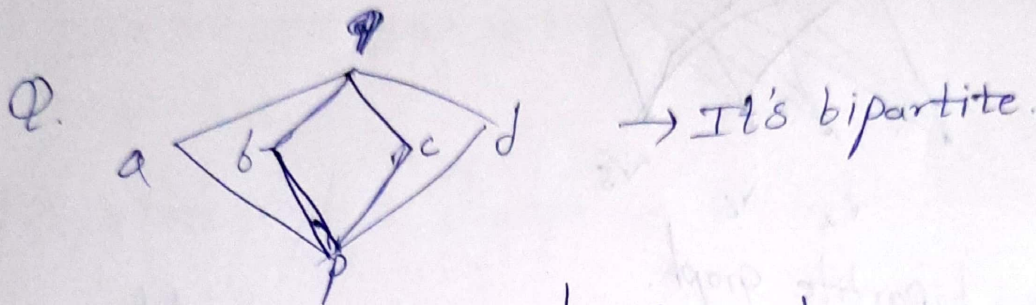
If G is simple, bipartite graph with m vertices & n edges then $m^2 \geq 4n$

→ Here $m=5$, $n=10$

$$5^2 \geq 4(10)$$

$$25 \not\geq 40$$

G_2 is not bipartite.



Complete Bipartite Graph: A graph $G(V, E)$ is called complete bipartite graph if:

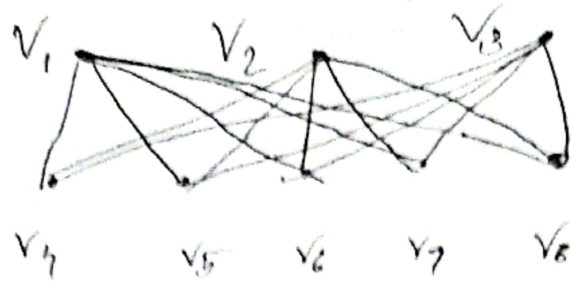
1) G is bipartite.

2) G is simple.

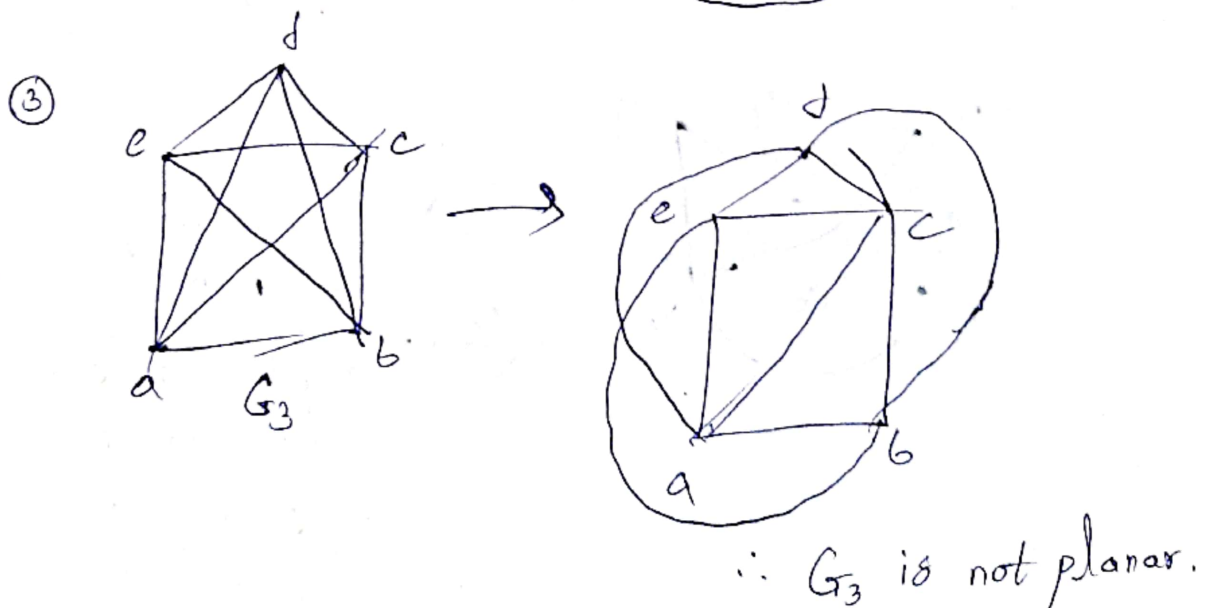
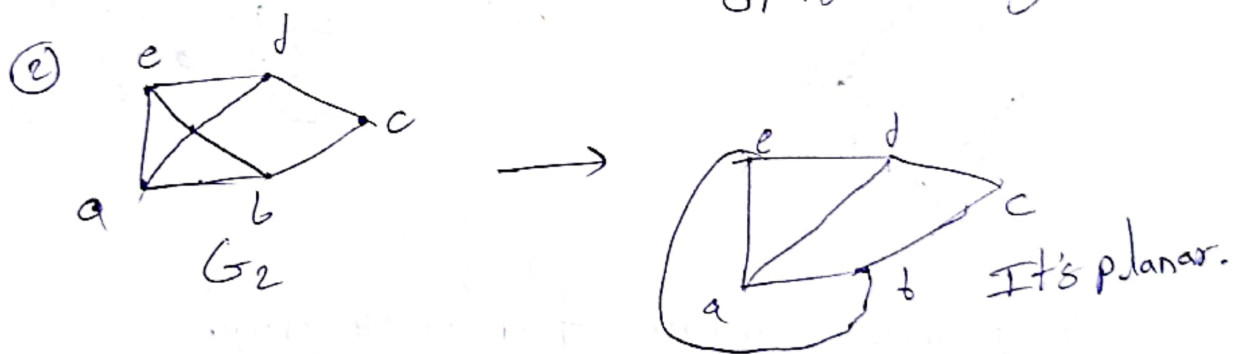
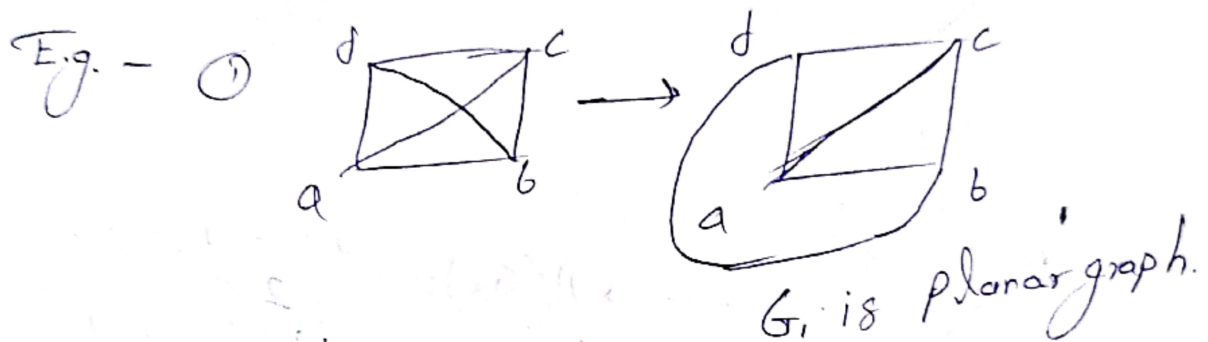
3) $V = U \cup W$ as partition of V then every vertex in U is adjacent to every vertex in W .
If U has m vertices & W has n vertices, complete bipartite graph is denoted by $K_{m,n}$.

E.g. - Let $U = \{v_1, v_2, v_3, \cancel{v_4}\}$
& $W = \{v_4, v_5, v_6, v_7, v_8\}$

$K_{3,5}$

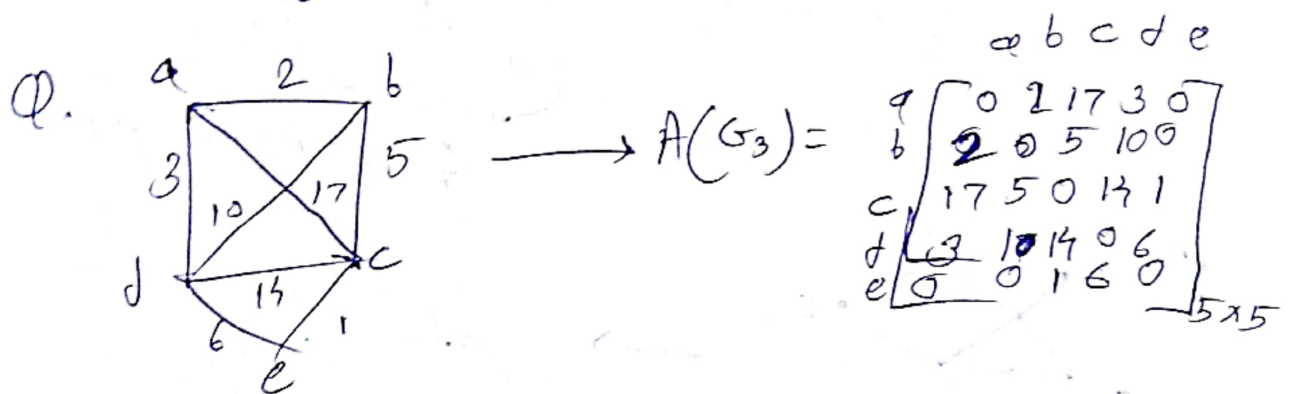
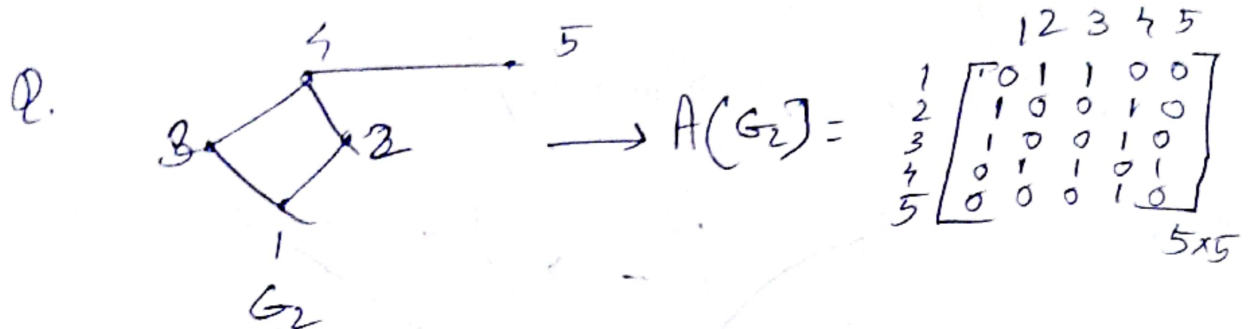
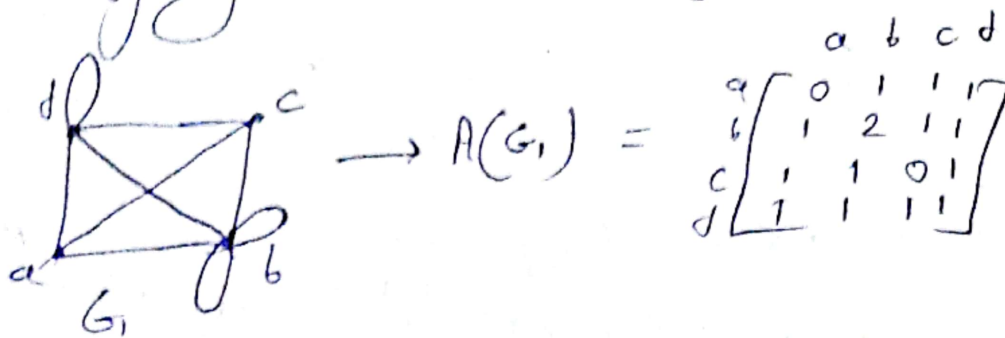


⑫ Planar Graph: It's possible to draw diagram of given graph in such a way that no two edges intersect then it's called a planar graph.

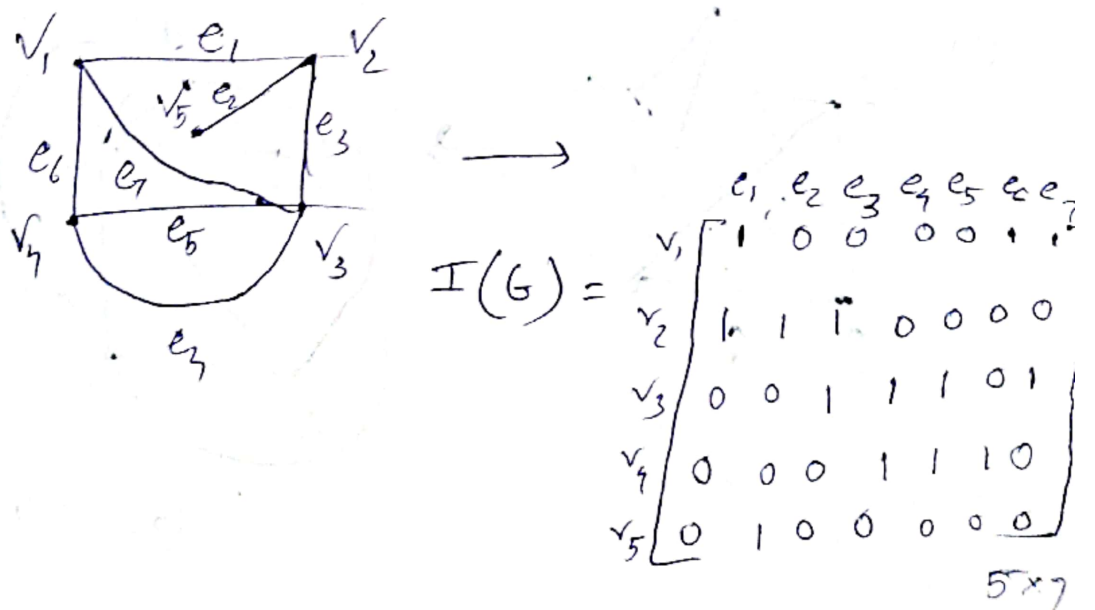


Matrix Representation of Graph:

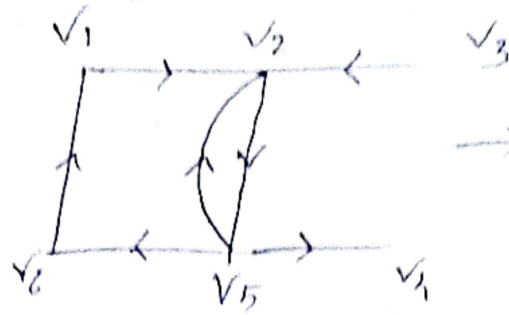
Q. Find adjacency matrix of following graph.



Q. Find incidence matrix of following graph.

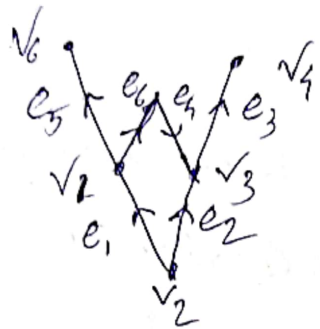


Q. Find adjacency matrix of following graph.



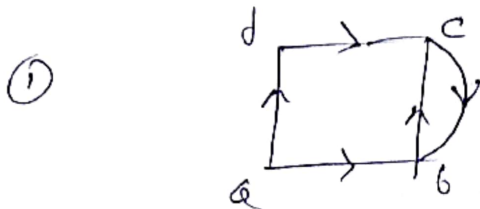
$$\rightarrow A(G) = \begin{matrix} & \begin{matrix} v_1 & v_2 & v_3 & v_4 & v_5 & v_6 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \\ v_6 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

Q. Find incidence matrix of following graph.

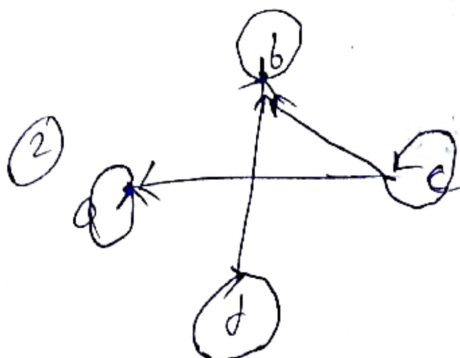


$$\rightarrow I(G) = \begin{matrix} & \begin{matrix} e_1 & e_2 & e_3 & e_4 & e_5 & e_6 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \\ v_6 \end{matrix} & \begin{bmatrix} -1 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 1 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 & -1 & 0 \end{bmatrix} \end{matrix}$$

Q. Find indegree & outdegree of following graph.

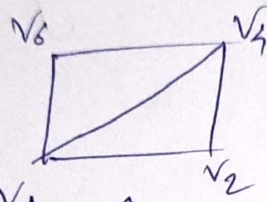
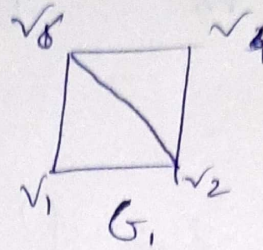
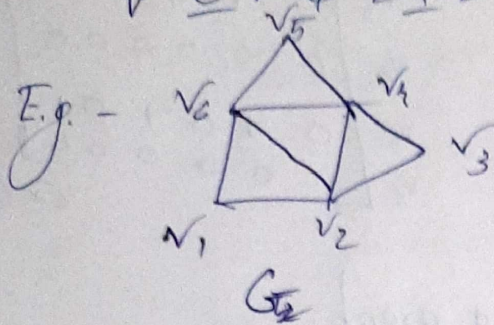


Vertex	In degree	Out degree
a	0	2
b	2	1
c	2	1
d	1	1



Vertex	In degree	Out degree
a	2	1
b	3	1
c	1	3
d	1	2

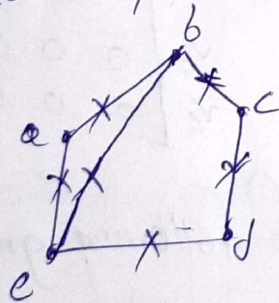
$\downarrow$ Subgraph: A graph G' is called a subgraph of graph G . If all vertices of G' are in G & all edges of G' are in G . i.e. if $G = (V, E)$ & $G' = (V', E')$
 $V' \subseteq V$ & $E' \subseteq E$.



$\rightarrow G_1$ is subgraph of G_2 $\swarrow$ G_2 is not a subgraph of G since v_1, v_4 is an edge in G_2 which isn't in G .

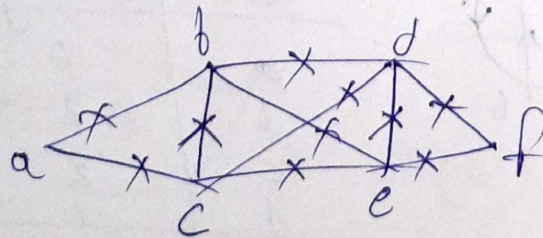
Q. Find Eulerian's path of following graph.

①



$\rightarrow$ Eulerian Path = $b c d e b a e$

②

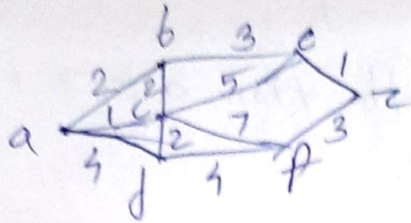


$\rightarrow b e f d c b d e c a b e$

1 Dijkstra's Algorithm:

Q. Find shortest path from A to z in the given graph using Dijkstra's Algorithm:

$$L(x) = \min \{old L(x), v(x) + w(v, x)\}$$



P \ T	a	b	c	d	e	f	z
a	0	∞	∞	∞	∞	∞	∞
a, c	-	2	1	4	∞	∞	∞
a, c, b	-	2	-	3	6	8	∞
a, c, b, e	-	-	-	3	5	8	∞
a, c, b, e, f	-	-	-	3	5	8	6

Step-① $T = \{a, b, c, d, e, f, z\}$

$$L(a) = 0 \quad \forall L(x) = \infty, \quad \forall x \in T, x \neq a$$

Step-② $L\{b\} = \min\{\text{old } L(b), v(a) + w(a, b)\}$

$$L\{b\} = \min\{\infty, 0 + 2\} = 2 \quad \checkmark$$

$$L\{c\} = \min\{\infty, 0 + 1\} = 1$$

$$L\{d\} = \min\{\infty, 0 + 4\} = 4$$

Step-③ $L\{b\} = \min\{2, 1 + 2\} = 2$

$$L\{d\} = \min\{4, 1 + 2\} = 3$$

$$L\{e\} = \min\{\infty, 1+5\} = 6$$

$$L\{f\} = \min\{\infty, 1+7\} = 8$$

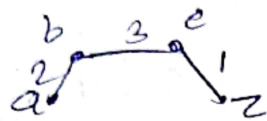
$$\text{step ④ } L\{e\} = \min\{6, 2+3\} = 5$$

$$\text{step ⑤ } L\{z\} = \min\{\infty, 5+1\} = 6$$

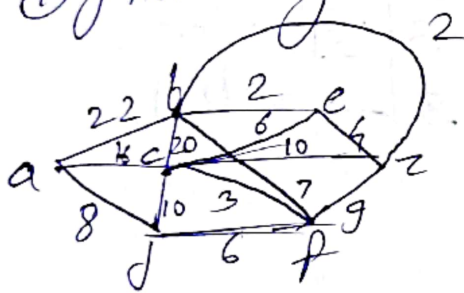
Hence length of shortest path from a to z is 6.

So, the shortest path is,

abez



Q. Find shortest path bet. ~~at~~ a to z by using ~~the~~ Dijkstra's algorithm.



→ step ① $T = \{a, b, c, d, e, f, z\}$

$$L(a) = 0 \text{ \& } L(x) = \infty, \forall x \in T, x \neq a$$

$$\text{step ②} - L\{b\} = \min\{\infty, 0+22\} = 22$$

$$L\{c\} = \min\{\infty, 0+16\} = 16$$

$$\checkmark L\{d\} = \min\{\infty, 0+8\} = 8$$

$$\text{step ③} - \cancel{L\{c\}} L\{c\} = \min\{16, 8+10\} = 16$$

$$\checkmark L\{f\} = \min\{\infty, 8+6\} = 14$$

±

$$\text{Step - (4)} - L\{c\} = \min\{16, 14+3\} = 16 \checkmark$$

$$L\{b\} = \min\{20, 14+7\} = 21$$

$$L\{z\} = \min\{\infty, 14+9\} = 23$$

$$\text{Step - (5)} - \checkmark L\{b\} = \min\{21, 16+20\} = 21$$

$$L\{c\} = \min\{\infty, 16+6\} = 22$$

$$L\{z\} = \min\{23, 16+10\} = 23$$

$$\text{Step - (6)} - L\{e\} = \min\{22, 21+2\} = 22 \checkmark$$

$$L\{z\} = \min\{23, 21+2\} = 23$$

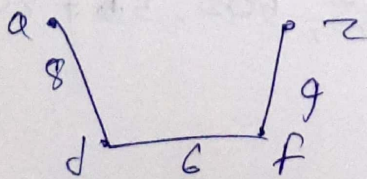
$$\text{Step - (7)} - L\{z\} = \min\{23, 22+1\} = 23$$

P \ T	a	b	c	d	e	f	z
a	0	∞	∞	∞	∞	∞	∞
b	-	22	16	8	∞	∞	∞
c	-	22	16	-	∞	14	∞
d	-	21	16	-	∞	-	23
e	-	21	-	-	22	-	23
f	-	-	-	-	22	-	23
z	-	-	-	-	-	-	23

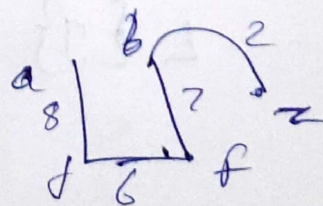
a to z

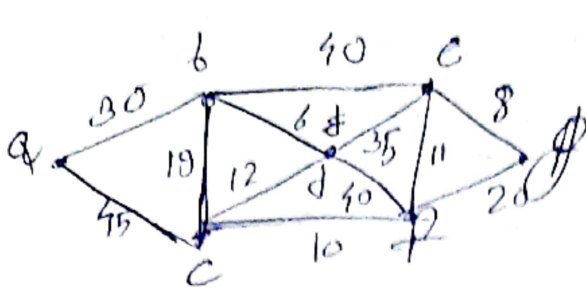
Hence length of shortest path is 23.

So, the shortest path is a d f z or a d f e z



OR





	a	b	c	d	e	f	g
a	0	30	45	∞	∞	∞	∞
b	30	0	40	12	19	∞	∞
c	45	40	0	35	11	∞	∞
d	∞	12	35	0	40	10	∞
e	∞	19	11	40	0	8	∞
f	∞	∞	∞	10	8	0	20
g	∞	∞	∞	∞	∞	20	0

Step ① - $T = \{a, b, c, d, e, f, g\}$

$$L(a) = 0 \text{ \& } L(x) = \infty \text{ if } x \in T, x \neq a$$

Step - ② - $L\{b\} = \min\{\infty, 0 + 30\} = 30 \checkmark$

$$L\{c\} = \min\{\infty, 0 + 45\} = 45$$

Step - ③ - $L\{c\} = \min\{45, 30 + 15\} = 45$

$$L\{e\} = \min\{\infty, 30 + 40\} = 70$$

$$L\{d\} = \min\{\infty, 30 + 6\} = 36 \checkmark$$

Step - ④ - $L\{e\} = \min\{70, 36 + 35\} = 70$

$$L\{f\} = \min\{\infty, 36 + 40\} = 76$$

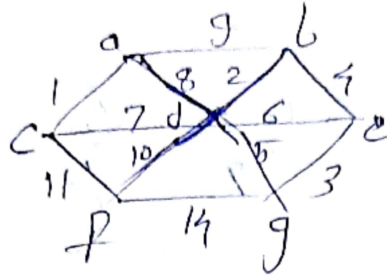
$$L\{c\} = \min\{45, 36 + 12\} = 48 \checkmark$$

Step - ⑤ - $L\{f\} = \min\{76, 48 + 10\} = 58$

Step - ⑥ - $L\{g\} = \min\{58, 58 + 20\} = 78$

f Kruskal's Algorithm:

Find minimum ~~span~~ spanning tree using Kruskal's algorithm
also find total cost of minimum spanning tree.



Sr. No.

Next edge to be added

Spanning Tree

1

{c,a}



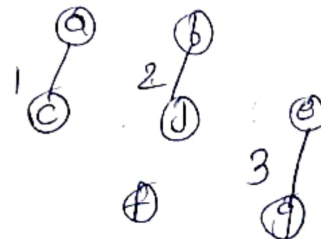
2

{b,f}



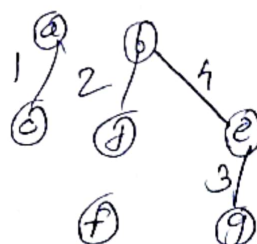
3

{e,g}



4

{b,e}

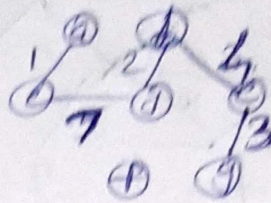


5

We can't add edge
dg bde because it'll
form cycle

6

cd

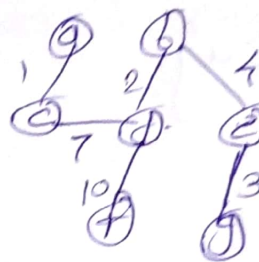


7

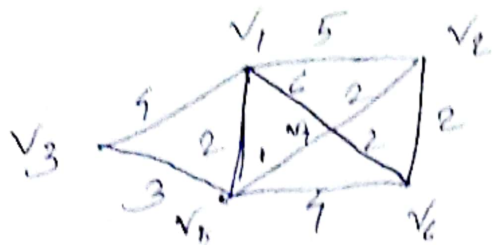
We can't add adf
ab because it'll
form cycle.

8

df



Total cost of minimum spanning tree = $1 + 2 + 3 + 4 + 7 + 10$
= 27



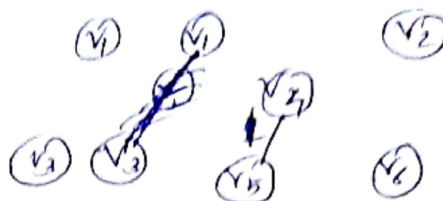
S. No.

Next edge to be added in spanning tree

Spanning tree

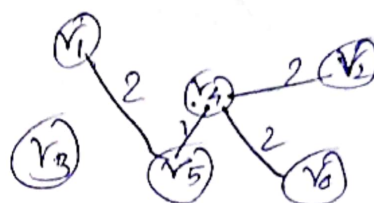
1.

v_4, v_5



2.

$(v_1, v_5), (v_4, v_6), (v_4, v_2)$

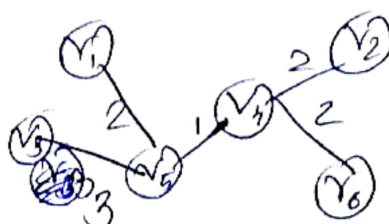
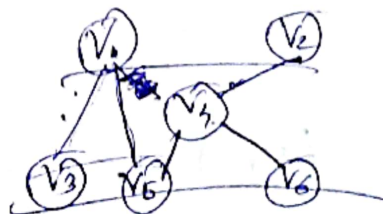


3.

We can't add v_2, v_6 because it will form cycle.

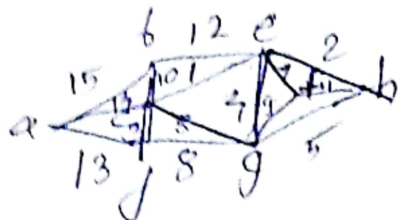
4.

v_3, v_5



$$1 + 2 + 2 + 2 + 3 = 10$$

④

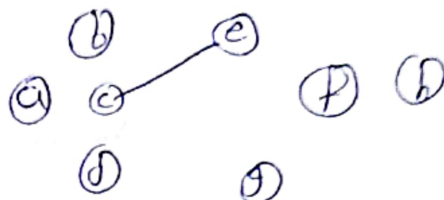


→ sr.no. Next edge to be added in Spanning tree

Spanning tree

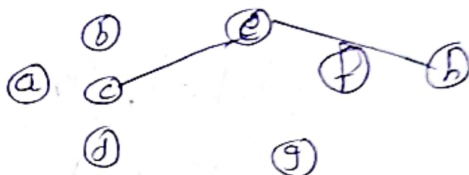
1.

(c,e)



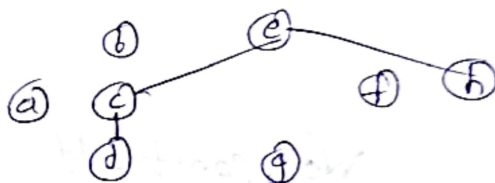
2.

(e,h)



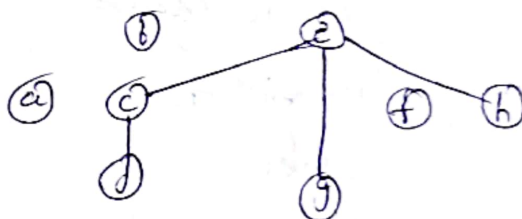
3.

(c,d)



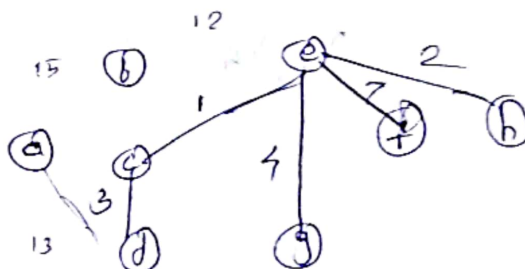
4.

(e,g)



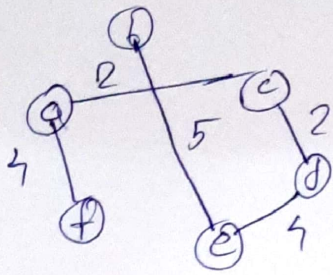
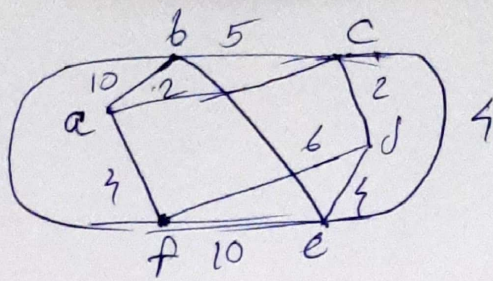
5.

(e,f)



6.

3



$$\rightarrow 4 + 2 + 5 + 2 + 4 = 17$$